Multimedia Signal processing

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1. **Question (8 pts)**:
   Consider a continuous source X, whose p.d.f. is given by

$$p_X(x) = \frac{1}{2} e^{-|x|} \tag{1}$$

   Let $\mathcal{Q}$ denote a scalar uniform quantizer with four quantization regions: $I_1 = (-\infty, -1]$, $I_2 = (-1, 0]$, $I_3 = (0, 1]$ and $I_4 = (+1, +\infty)$.
   (a) Compute the entropy, in bits / symbol, of the source representing the output of the quantizer.
   (b) Find the optimal values of the reconstruction levels that minimize the MSE distortion.
   (c) Compute the MSE distortion of the quantizer above (you are not requested to perform the computations exhaustively).
   Hint. You might find the following indefinite integral useful

$$\int a e^{ca} da = \frac{e^{ca}}{c}(ca - 1) \tag{2}$$

$$\int a^2 e^{ca} da = e^{ca}\left(\frac{a^2}{c} - \frac{2a}{c^2} + \frac{2}{c^3}\right) \tag{3}$$

2. **Question (5 pts)**
   Discuss the design of the perceptual model in audio coding.
3. **Question (5 pts)**
   (a) Illustrate by means of a pseudocode, motion estimation by block matching, clearly identifying the input and the output (both data and parameters).
   (b) Discuss the computational complexity of block matching and the measures that can be adopted to reduce it.

4. **Questions (5 pts - each answer can be either TRUE or FALSE)**
   (a) Consider a transform coding system that processes blocks of N input samples.
   - [T] [F] The variances of the transform coefficients $\sigma^2_{y_k}$, $k = 1, \ldots, N$, are always different from each other.
   - [T] [F] When the optimal transform is used, the transform coefficients are uncorrelated.
   - [T] [F] $R_{TC}(D) \leq R_{VQ}(D)$, where $R_{TC}(D)$ and $R_{VQ}(D)$ are, respectively, the operational rate-distortion curves of transform coding and vector quantization.
   - [T] [F] The number of bits allocated to each transform coefficient is $R_k \propto \log_2 \sigma^2_{y_k}$

   (b) Consider lossy coding a source without memory
   - [T] [F] $D_{LM}(R) \leq D_U(R)$, where LM and U refer to, respectively, the Lloyd-Max and uniform quantizer.
   - [T] [F] In a Lloyd-Max quantizer, the reconstruction levels are equally likely.
   - [T] [F] $D_{VQ}(R) = D_{SQ}(R)$, where VQ and SQ denote, respectively, vector and scalar quantization.

   (c) Consider a video coding system.
   - [T] [F] Motion vectors are lossy coded.
   - [T] [F] The average PSNR of prediction residuals increases when increasing the size of the block used in motion estimation/compensation.
   - [T] [F] Temporal redundancy is addressed by means of DPCM.

5. **Question (Matlab - 10pts)** Consider the images *image1.png* and *image2.png*.
   For each of them
   (a) Compute the block-wise DCT with blocks of size $N \times N$.
   (b) Compute the block-wise KLT with blocks of size $N \times N$.
   (c) Estimate the transform coding gain in three cases, when $N = 8$.
   - DCT
   - KLT trained on the same image
   - KLT trained on the other image

   (d) Show the result as a bar chart and comment.
   (e) Let N vary in the set $\{4, 8, 16, 32\}$, and comment the result.